Supervised Functional Data Discriminant Analysis for Hyperspectral Image Classification

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***Abstract*— This article proposes a functional data discriminant analysis (FDDA) method for hyperspectral image (HSI) classification. This method analyzes and processes the HSI data from a functional point of view, which is a novel perspective in HSI processing. The classical methods achieve dimensionality reduction by directly eliminating the redundancy of the HSI data. However, the proposed method extracts the functional features by utilizing the redundancy of the HSI data. Functional features can effectively reveal inherent characteristics of the HSI data with the change in the wavelengths. Based on this, a regularized weighted fitting model is first built for converting a spectral vector into a spectral curve. Second, an FDDA method defined in the function field is presented for extracting the functional features of the spectral curves. Finally, a novel spectral–spatial framework is designed for classification tasks of HSI data sets. Experimental results in three commonly used HSI data sets indicate that the proposed method is effective and leads to promising classification results compared with some benchmarking methods. More importantly, the work tries to diversify and develop the existing theory and methods of HSI classification from discrete (vector) data learning methods to continuous (functional) data learning methods.**

***Index Terms*— B-spline basis function, functional data discriminant analysis (FDDA), hyperspectral image (HSI) classification, regularized weighted fitting model (RWFM).**

I. Introduction

HYPERSPECTRAL sensors often capture data from the visible through the near-infrared wavelength ranges. With the spectral resolution increasing, the hyperspectral image (HSI) data generally records a nearly continuous spectral information for land covers. The rich spectral information provides the possibility for the accurate classification of land covers. However, HSI classification is very challenging because of the limited number of available labeled samples, the presence of redundant features, and the high dimensionality of the data [1].

According to the availability of labeled samples, the commonly used classification methods are mainly divided into three categories, i.e., unsupervised, supervised, and semisupervised approaches. Unsupervised classification can implement the classification task without labeled samples. Although unsupervised methods are relatively simple and feasible, their classification is usually not very accurate but can be useful for preprocessing, i.e., prior to supervised classification. In HSI classification, the common methods mainly include fuzzy C-means [2], self-organizing maps (SOMs) [3], and unsupervised artificial immune classifier (UAIC) [4]. In contrast, supervised approaches classify unknown/test samples based on labeled samples, which are the most widely researched methods in HSI classification. Moreover, the classification accuracy of supervised approaches is generally better than that of unsupervised ones. Up until now, many kinds of supervised methods have been presented, such as k-nearest neighbor (K-NN) [5], support vector machines (SVMs) [6]–[10], random forest (RF) [11], [12], multinomial logistic regression (MLR) [13], and deep learning methods [14]–[16]. Besides unsupervised and supervised methods, semisupervised methods also catch many researchers' attention. Due to the joint use of labeled and unlabeled samples in the processing of sample training, semisupervised methods can generally outperform the former two methods in terms of classification accuracies. However, semisupervised learning is much more costly in terms of computations. In recent years, various semisupervised methods have been proposed that have achieved very good classification results [17]–[21]. This article focuses on supervised classification.

In the case of insufficient labeled samples, the high dimensionality of the spectral vectors and the presence of redundant information brings a great difficulty for the precise classification of land covers. Feature extraction can effectively solve this problem. It aims at extracting the most effective features of the original data by reducing the dimensionality and eliminating the redundant features, which is one of the most important research topics in classification. Many common

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feature extraction methods have been introduced, ranging from linear approaches to nonlinear ones, such as principal component analysis (PCA) and its kernel method (KPCA) [22]–[25], linear discriminant analysis (LDA) [26], locally linear embedding (LLE) [27]–[29], nonparametric weighted feature extraction (NWFE) [30], and deep feature extraction methods [31], [32]. All these methods are designed to reduce the dimensionality by eliminating the redundancy of the HSI data. This means some information in the original data will be lost. The error arising from the information loss in the original data can affect the discriminative power of classifiers. It is very difficult to determine whether these methods can accurately extract inherent features of the data. The reason is that most existing methods only regard the spectral information as vectors of reflectance values without considering their band ordering. According to the HSI principle, spectral information is captured with a very fine spectral resolution, giving the appearance of continuous curves. Moreover, with the spectral resolution increasing, the spectral information gradually forms complete and continuous spectral curves as a function of the wavelengths. Thus, the essence of the spectral information is the functional data [33]. In this article, we analyze the HSI data from a functional point of view and attempt to search effective methods for the accurate classification of land covers.

Functional data analysis provides very good tools for high-dimensional functional data [33], [34]. Based on this, a few functional data classification methods have been proposed in some fields [35], [36]. In HSI processing, some researchers also make their attempt and introduce the unsupervised functional PCA (FPCA) for HSI classification. For instance, in our previous works, Li *et al.* [37] introduced the FPCA to extract the functional features of spectral information and classified them with SVM. Later, Wei *et al.* [38] also introduced the FPCA to extract the functional features of HSIs and used the kernel extreme learning machine (KELM) for classification. Recently, Majdar *et al.* [39] proposed a spectral–spatial classification method using FPCA and a watershed segmentation algorithm. Lv *et al.* [40] also proposed a spatial FPCA method for HSI classification. Furthermore, Zullo *et al.* [41] proposed the functional extension of SVM and RF based on FPCA for HSI classification. These methods achieved very good accuracies. However, these works were based on the FPCA only. The methodology of functional data analysis for HSI processing needs to be further developed. New methods need to be proposed for enriching the existing theory and algorithms in HSI classification.

Based on our previous works, this article proposes a supervised functional data discriminant analysis (FDDA) method for HSI classification. First, by considering the continuity of spectral curves, a basis function system is designed to fit the spectral information. Then, the functional representations can be obtained from the spectral information by a regularized weighted fitting model (RWFM). According to functional representations, a functional feature extraction method is built from a functional point of view. Simultaneously, a fixed neighborhood method is designed to extract the spatial information. This method can keep the functional characteristic of the spatial information. Furthermore, the functional spatial features are extracted by the proposed functional feature extraction method. Finally, the functional spectral and spatial features are fused. SVM is used for the classification task of the fused features. The proposed functional feature extraction method can effectively handle the problems of the high dimensionality, strong correlation, and redundancy of the HSI data. Furthermore, this work tries to develop the existing theory and algorithms for HSI classification by combining the various modes of data processing including the vector data and functional data processing. The main contributions of this article consist of the following.

1) An RWFM is proposed for converting a spectral vector into a spectral curve. The proposed model can obtain a very good fit and not only can effectively overcome the noise but also ensure the smoothness of curves.
2) The FDDA method in the function space is built for feature extraction of infinite-dimensional curves. This method aims at improving the similarity between spectral curves from the same class and highlighting the differences between ones from different classes by seeking an optimal eigenfunction.

This article is organized into five sections. Section II discusses the basis function systems, and RWFM. Section III presents the supervised functional feature extraction method for spectral curves and gives a spectral–spatial classification framework of the proposed algorithm. Section IV shows the experimental results on three HSI data sets and the contrastive analysis between the proposed method and some benchmarking methods. Finally, the conclusions are drawn in Section V.

II. Functional Data Fitting Model

According to the HSI principle, each pixel of the HSI has tens or hundreds of magnitude values corresponding to its wave bands. The wave bands are sampled at intervals from the visible through the near-infrared wavelength ranges. With the increase in spectral resolution, each pixel can form a complete and continuous spectral curve. Based on this, we can see that the HSI data are the functional data. Thus, it is a novel idea to convert the HSI data into spectral curves and process them from the function point of view.

A. Basis Function Systems

This section serves to introduce how to turn the raw discrete data into continuous functional curves. One of the key technologies is the basis function representation method. The spectral vector of each pixel can be regarded as sampling points in a functional curve. Thus, we want to fit these sampling points into a curve. Different pixels have different curves. With that in mind, the first step is to consider the spectral information as the functional data.

Assume that we have a HSI $\boldsymbol{X} \in \mathbb{R}^{N\times m}$ consisting of N pixels $\{\boldsymbol{x}_i, y_i\}_{i=1}^{N}$, where $\boldsymbol{X} = \{\boldsymbol{x}_1, \ldots, \boldsymbol{x}_N\}$. Here, $\boldsymbol{x}_i \in \mathbb{R}^m$ is any spectral vector, and y_i is the corresponding label in the class label set $\boldsymbol{\Omega} = \{1, \ldots, C\}$. Furthermore, the m elements in the spectral vector $\boldsymbol{x}_i = \{\boldsymbol{x}_{i1}, \ldots, \boldsymbol{x}_{im}\}$ are acquired in sequence from the corresponding spectral curve $x_i(t)$, where t denotes the wavelength. Then, the jth element in $\boldsymbol{x}_i$ can be

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transformed into a point at the wavelength t_j on the curve $x_i(t)$, that is,

$$\boldsymbol{x}_{ij} = x_i(t_j), \quad j = 1, 2, \ldots, m. \tag{1}$$

By means of (1), the raw discrete data are transformed into the functional data. Next, we use the basis function system to fit the function data. Because HSIs generally contain noise, the functional data fitting model is very suitable to process HSIs. The main idea is to denote the spectral curve $x_i(t)$ as a linear combination of a basis function system with the noise, that is,

$$x_i(t) = \sum_{k=1}^{K} c_{ik}\phi_k(t) + \varepsilon_i(t) \tag{2}$$

where $\varepsilon_i(t)$ is the noise function. Here, $\{c_{ik}\}_{k=1}^{K}$ is the functional representations of $x_i(t)$ under the basis function system $\{\phi_k(t)\}_{k=1}^{K}$. The basis function system $\{\phi_k(t)\}_{k=1}^{K}$ is a set of known functions that are mathematically independent of each other. Moreover, the basis function system has the property that it can approximate arbitrarily well any function by taking a weighted sum or linear combination of a sufficiently large number K of the basis functions [33]. The common basis function systems mainly include the polynomial basis system, the Fourier basis system, the B-spline basis system, the wavelet basis system, and so on. According to the property of basis functions, the corresponding basis function system has its range of application. For example, the Fourier basis system is suitable for periodic data, while the B-spline basis system is suitable for open-ended data. With that in mind, the B-spline basis system is a well good choice for HSI processing.

Let us define any pixel $\boldsymbol{x}_i = \{x_i(t_1), \ldots, x_i(t_m)\}$, $t_1 < \cdots < t_m$. In order to define the B-spline basis system, the order p of the B-spline basis functions needs to be introduced. The notation $B_{k,j}(t)$ is used to denote the kth B-spline basis function of the order $j (j = 1, \ldots, p)$. Thus, the B-spline basis system can be defined as [33]

$$\begin{aligned} B_{k,1}(t) &= \begin{cases} 1, & \text{if } t_k \le t < t_{k+1} \\ 0, & \text{otherwise} \end{cases}, \quad k = 1, \ldots, m-1 \\ B_{k,j}(t) &= \frac{t - t_k}{t_{k+j-1} - t_k} B_{k,j-1}(t) + \frac{t_{k+j} - t}{t_{k+j} - t_{k+1}} B_{k+1,j-1}(t). \end{aligned} \tag{3}$$

In the latter equation above, $j = 2, 3, \ldots, p$, and $k = 1, 2, \ldots, m - j$. The number of the basis function K can uniquely be determined by the B-spline order p and the band number m, i.e., $K = m - p$. The B-spline basis functions are smooth and well-behaved when p is not less than 2. Moreover, increasing p does not always improve the effect of data fitting. Thus, p is usually set relatively small to reduce the computational cost. In general, p is set to 3. The spectral curve $x_i(t)$ of any pixel $\boldsymbol{x}_i$ can be denoted as a linear combination of the above B-spline basis system

$$\begin{aligned} x_i(t) &= \sum_{k=1}^{K} c_{ik} B_{k,p}(t) + \varepsilon_i(t) \\ &= \boldsymbol{B}(t)^T \boldsymbol{c}_i + \varepsilon_i(t) = \boldsymbol{c}_i^T \boldsymbol{B}(t) + \varepsilon_i(t) \end{aligned} \tag{4}$$

where $\boldsymbol{B}(t) = (B_{1,p}(t), B_{2,p}(t), \ldots, B_{K,p}(t))^T$, and $\boldsymbol{c}_i = (c_{i1}, c_{i2}, \ldots, c_{iK})^T$. Next, we focus on building the functional data fitting model by minimizing the sum of squared errors.

B. Regularized Weighted Fitting Model

This section serves to introduce how to turn discrete functional data into smooth functions. According to (4), a simple least squares fitting model can be built by minimizing the sum of squared errors

$$\begin{aligned} S_e(i) &= \sum_{j=1}^{m} \varepsilon_i(t_j)^2 = \sum_{j=1}^{m} \left[x_i(t_j) - \sum_{k=1}^{K} c_{ik} B_{k,p}(t_j) \right]^2 \\ &= (\boldsymbol{x}_i - \boldsymbol{B}\boldsymbol{c}_i)^T (\boldsymbol{x}_i - \boldsymbol{B}\boldsymbol{c}_i) \end{aligned} \tag{5}$$

where $\boldsymbol{x}_i = (x_i(t_1), x_i(t_2), \ldots, x_i(t_m))^T$, and

$$\boldsymbol{B} = \begin{pmatrix} \boldsymbol{B}(t_1) \\ \vdots \\ \boldsymbol{B}(t_m) \end{pmatrix} = \begin{pmatrix} B_{1,p}(t_1) & \cdots & B_{K,p}(t_1) \\ \vdots & \ddots & \vdots \\ B_{1,p}(t_m) & \cdots & B_{K,p}(t_m) \end{pmatrix}. \tag{6}$$

Minimizing the term in (5) can easily be solved by the least squares method. The simple least squares fitting model is appropriate in situations where it can be assumed that the noise $\varepsilon_i(t)$ is independent and identically distributed, described by the Gaussian distribution with the mean 0, and the variance σ^2. In fact, the noise $\varepsilon_i(t)$ does not satisfy this ideal condition. To deal with nonstationary errors, we need to build a weighted fitting model by extending (5)

$$\begin{aligned} \mathrm{WS}_e(i) &= \sum_{j=1}^{m} \omega_i(t_j) \varepsilon_i(t_j)^2 \\ &= \sum_{j=1}^{m} \omega_i(t_j) [x_i(t_j) - \sum_{k=1}^{K} c_{ik} B_{k,p}(t_j)]^2 \\ &= (\boldsymbol{x}_i - \boldsymbol{B}\boldsymbol{c}_i)^T \boldsymbol{W}_i (\boldsymbol{x}_i - \boldsymbol{B}\boldsymbol{c}_i) \end{aligned} \tag{7}$$

where $\boldsymbol{W}_i = \mathrm{diag}\{\omega_i(t_1), \omega_i(t_2), \ldots, \omega_i(t_m)\}$ is a diagonal matrix, and the weight value $\omega_i(t_j)$ is related with the error $\varepsilon_i(t_j)$. The larger the error is, the smaller the weight is. Thus, the weight value $\omega_i(t_j)$ can be constructed by using the following kernel function:

$$\omega_i(t_j) = exp\left(-\frac{\varepsilon_i(t_j)^2}{h^2} \right) \tag{8}$$

where h is the bandwidth parameter. This weighted fitting model can effectively avoid the disturbance of the noise or outliers. In fact, the error $\varepsilon_i(t_j)$ is prior unknown. Instead of the real error, its estimator $\hat{\varepsilon}_i(t_j)$ is computed by data points near t_j and using localized least squares. The basis expansions can provide good approximations to the HSI data. However, the weighted fitting model cannot offer continuous control over the degree of smoothing. To ensure the smoothness of the spectral curve $x_i(t)$, an RWFM is built by introducing the

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regularization term

$$\begin{aligned}\mathrm{RWS}_e(i) &= \sum_{j=1}^{m}\omega_i(t_j)\varepsilon_i(t_j)^2 + \lambda\int[D^2x_i(t)]^2dt \\ &= (\boldsymbol{x}_i - \boldsymbol{B}\boldsymbol{c}_i)^T\boldsymbol{W}_i(\boldsymbol{x}_i - \boldsymbol{B}\boldsymbol{c}_i) + \lambda\int[D^2x_i(t)]^2dt\end{aligned} \tag{9}$$

where $\lambda(\lambda \geq 0)$ is a smoothing parameter, which is used for balancing the error and the smoothness. The smoothing parameter, λ, tends to be small and positive. The term $D^2x_i(t)$ represents the second derivative of a function $x_i(t)$ and $[D^2x_i(t)]^2$ aims at quantifying the smoothness of $x_i(t)$. In general, the smaller $[D^2x_i(t)]^2$ is, the smoother the curve $x_i(t)$ is at t. Without doubt, utilizing the higher derivative can improve the smoothness of $x_i(t)$, but increase the computational cost. In fact, the function $x_i(t)$ is unknown, and its second derivative $D^2x_i(t)$ cannot be directly computed. Thus, the function $x_i(t)$ in the regularization term is approximated by the linear combination of the B-spline basis system. The regularization term in (9) can be rewritten as

$$\begin{aligned}&\int[D^2x_i(t)]^2dt \\ &= \int\left[D^2\left(\sum_{k=1}^{K}c_{ik}B_{k,p}(t)\right)\right]^2dt \\ &= \int\left[\sum_{k=1}^{K}c_{ik}D^2B_{k,p}(t)\right]^2dt \\ &= \int\left[\sum_{k=1}^{K}c_{ik}D^2B_{k,p}(t)\right]\left[\sum_{h=1}^{K}c_{ih}D^2B_{h,p}(t)\right]dt \\ &= \sum_{k=1}^{K}\sum_{h=1}^{K}c_{ik}\left[\int D^2B_{k,p}(t)D^2B_{h,p}(t)dt\right]c_{ih} \\ &= \boldsymbol{c}_i^T\boldsymbol{R}\boldsymbol{c}_i.\end{aligned} \tag{10}$$

Here, $\boldsymbol{R}$ is the correlation matrix of the second derivative of the B-spline basis system, which is defined by the inner product of functions, that is,

$$< f, g >= \int f(t)g(t)dt. \tag{11}$$

Thus, the element $\boldsymbol{R}(i, j)$ of the matrix $\boldsymbol{R}$ in the ith row and jth column can be computed by

$$\boldsymbol{R}(k, h) =< D^2B_{k,p}, D^2B_{h,p} > . \tag{12}$$

According to (10), (9) can be simplified as

$$\begin{aligned}\mathrm{RWS}_e(i) &= (\boldsymbol{x}_i - \boldsymbol{B}\boldsymbol{c}_i)^T\boldsymbol{W}_i(\boldsymbol{x}_i - \boldsymbol{B}\boldsymbol{c}_i) + \lambda\int[D^2x_i(t)]^2dt \\ &= (\boldsymbol{x}_i - \boldsymbol{B}\boldsymbol{c}_i)^T\boldsymbol{W}_i(\boldsymbol{x}_i - \boldsymbol{B}\boldsymbol{c}_i) + \lambda\boldsymbol{c}_i^T\boldsymbol{R}\boldsymbol{c}_i.\end{aligned} \tag{13}$$

Minimizing (13) is a classic unconstrained optimization problem. In general, it is easily solved by taking the partial derivative with respect to the parameter vector $\boldsymbol{c}_i$. We can obtain the solution, that is,

$$\hat{\boldsymbol{c}}_i = (\boldsymbol{B}^T\boldsymbol{W}_i\boldsymbol{B} + \lambda\boldsymbol{R})^{-1}\boldsymbol{B}\boldsymbol{W}_i\boldsymbol{x}_i. \tag{14}$$

According to the RWFM, each pixel $\boldsymbol{x}_i$ in a HSI can be fit to a functional curve, $x_i(t)$

$$x_i(t) = \boldsymbol{B}(t)^T\hat{\boldsymbol{c}}_i = \boldsymbol{B}(t)^T(\boldsymbol{B}^T\boldsymbol{W}_i\boldsymbol{B} + \lambda\boldsymbol{R})^{-1}\boldsymbol{B}\boldsymbol{W}_i\boldsymbol{x}_i. \tag{15}$$

Based on the resulting spectral curves $\{x_i(t)\}_{i=1}^n$ and their corresponding labels $\{y_i\}_{i=1}^n$, we consider building a supervised functional feature extraction method in the function space.

III. Supervised Functional Feature Extraction Method

In this section, we discuss the classification task for functional curves. To solve this infinite-dimensional problem, we need to build an effective functional feature extraction method to improve the similarity between spectral curves from the same class and highlight the differences between ones from different classes. In the classical multivariate case, LDA can achieve this goal. Its main idea is to seek the optimal projection direction on which the spectral vectors between different classes are far from each other and in the same class to be close to each other [26]. With that in mind, we build an FDDA method in the function space.

A. Functional Data Discriminant Analysis

Assuming that n labeled samples $\{\boldsymbol{x}_i, y_i\}_{i=1}^n$ in a HSI are given, the objective function of LDA is

$$\max_{\boldsymbol{w}} J(\boldsymbol{w}) = \frac{\boldsymbol{w}^T\boldsymbol{S}_b\boldsymbol{w}}{\boldsymbol{w}^T\boldsymbol{S}_t\boldsymbol{w}} \tag{16}$$

where $\boldsymbol{S}_b = \sum_{k=1}^{C}(\boldsymbol{\mu}_k - \boldsymbol{\mu})(\boldsymbol{\mu}_k - \boldsymbol{\mu})^T$ and $\boldsymbol{S}_t = \sum_{i=1}^{n}(\boldsymbol{x}_i - \boldsymbol{\mu})(\boldsymbol{x}_i - \boldsymbol{\mu})^T$ are the between-class and total scatter matrices, respectively. Here, $\boldsymbol{\mu} = (1/n)\sum_{i=1}^{n}\boldsymbol{x}_i$ is the global mean, $\boldsymbol{\mu}_k = (1/n_k)\sum_{j=1}^{n_k}\boldsymbol{x}_j^k$ is the mean of the kth class, n_k is the number of samples in the kth class, $n = \sum_{k=1}^{C}n_k$ and $\boldsymbol{x}_j^k$ is the jth sample in the k-th class. Next, we build the similar objective function in the function space. The total scatter matrix can be seen as calculated by some discrete points $\boldsymbol{x}_i = \{\boldsymbol{x}_{i1}, \boldsymbol{x}_{i2}, \ldots, \boldsymbol{x}_{im}\}$ sampled from the spectral curves $\{x_i(t)\}_{i=1}^n$. Thus, the total scatter matrix can be transformed back into the functional form, which is defined as the total scatter function about t and s

$$S_t(t, s) = \sum_{i=1}^{n}(x_i(t) - \mu(t))(x_i(s) - \mu(s)) \tag{17}$$

where $\mu(t) = (1/n)\sum_{i=1}^{n}x_i(t)$ is the global mean curve, which can be transformed as the vector form according to (15)

$$\mu(t) = \frac{1}{n}\sum_{i=1}^{n}x_i(t) = \frac{1}{n}\sum_{i=1}^{n}\boldsymbol{B}(t)^T\hat{\boldsymbol{c}}_i = \boldsymbol{B}(t)^T\boldsymbol{\mu}_c = \boldsymbol{\mu}_c^T\boldsymbol{B}(t) \tag{18}$$

where $\boldsymbol{\mu}_c = (1/n)\sum_{i=1}^{n}\hat{\boldsymbol{c}}_i$ is a K-dimensional vector, which is the mean of functional representations of spectral curves

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from all labeled samples. Thus, (17) can be rewritten as

$$\begin{aligned} S_t(t,s) &= \sum_{i=1}^{n}(x_i(t)-\mu(t))(x_i(s)-\mu(s)) \\ &= \sum_{i=1}^{n}(\boldsymbol{B}(t)^T\hat{\boldsymbol{c}}_i-\boldsymbol{B}(t)^T\boldsymbol{\mu}_c)\left(\hat{\boldsymbol{c}}_i^T\boldsymbol{B}(s)-\boldsymbol{\mu}_c^T\boldsymbol{B}(s)\right) \\ &= \boldsymbol{B}(t)^T\left[\sum_{i=1}^{n}(\hat{\boldsymbol{c}}_i-\boldsymbol{\mu}_c)(\hat{\boldsymbol{c}}_i-\boldsymbol{\mu}_c)^T\right]\boldsymbol{B}(s) \\ &= \boldsymbol{B}(t)^T\boldsymbol{S}_t^c\boldsymbol{B}(s). \end{aligned} \tag{19}$$

Here, $\boldsymbol{S}_t^c=\sum_{i=1}^{n}(\hat{\boldsymbol{c}}_i-\boldsymbol{\mu}_c)(\hat{\boldsymbol{c}}_i-\boldsymbol{\mu}_c)^T$ is the total scatter matrix of functional representation vectors $\{\hat{\boldsymbol{c}}_i\}_{i=1}^n$. Similarly, we can define the between-class scatter function about t and s

$$\begin{aligned} S_b(t,s) &= \sum_{k=1}^{C}(\mu_k(t)-\mu(t))(\mu_k(s)-\mu(s)) \\ &= \sum_{k=1}^{C}\left(\boldsymbol{B}(t)^T\boldsymbol{\mu}_c^k-\boldsymbol{B}(t)^T\boldsymbol{\mu}_c\right)\left(\boldsymbol{\mu}_c^{kT}\boldsymbol{B}(s)-\boldsymbol{\mu}_c^T\boldsymbol{B}(s)\right) \\ &= \boldsymbol{B}(t)^T\left[\sum_{k=1}^{C}\left(\boldsymbol{\mu}_c^k-\boldsymbol{\mu}_c\right)\left(\boldsymbol{\mu}_c^k-\boldsymbol{\mu}_c\right)^T\right]\boldsymbol{B}(s) \\ &= \boldsymbol{B}(t)^T\boldsymbol{S}_b^c\boldsymbol{B}(s) \end{aligned} \tag{20}$$

where $\boldsymbol{S}_b^c=\sum_{k=1}^{C}(\boldsymbol{\mu}_c^k-\boldsymbol{\mu}_c)(\boldsymbol{\mu}_c^k-\boldsymbol{\mu}_c)^T$ is the between-class scatter matrix of functional representation vectors $\{\hat{\boldsymbol{c}}_i\}_{i=1}^n$. Here, $\mu_k(t)$ is the mean curve of the kth class, which is denoted as

$$\begin{aligned} \mu_k(t) &= \frac{1}{n_k}\sum_{j=1}^{n_k}x_j^k(t)=\frac{1}{n_k}\sum_{j=1}^{n_k}\boldsymbol{B}(t)^T\hat{\boldsymbol{c}}_j^k \\ &= \boldsymbol{B}(t)^T\boldsymbol{\mu}_c^k=\boldsymbol{\mu}_c^{kT}\boldsymbol{B}(t) \end{aligned} \tag{21}$$

where $x_j^k(t)$ is the spectral curve of the jth sample $\boldsymbol{x}_j^k$, $\hat{\boldsymbol{c}}_j^k$ is the corresponding functional representations, and $\boldsymbol{\mu}_c^k=(1/n_k)\sum_{j=1}^{n_k}\hat{\boldsymbol{c}}_j^k$ is the mean of functional representations in the kth class. Next, we want to seek an optimal integral transform function ξ like the optimal projection direction in LDA. It aims at measuring the scatter degree in the sense of integration. Thus, we can build the objective function of FDDA in the function space by double integrals about t and s

$$\max_{\xi}\ J_f(\xi)=\frac{\int\int\xi(t)S_b(t,s)\xi(s)dtds}{\int\int\xi(t)S_t(t,s)\xi(s)dtds}. \tag{22}$$

From the Riesz representation theorem [42], we know that any solution ξ must lie in the Hilbert space $\mathcal{H}$ spanned by the B-spline basis system. Therefore, we can get

$$\xi(t)=\sum_{k=1}^{K}\alpha_k B_{k,p}(t)=\boldsymbol{\alpha}^T\boldsymbol{B}(t)=\boldsymbol{B}(t)^T\boldsymbol{\alpha} \tag{23}$$

where $\boldsymbol{\alpha}=(\alpha_1,\alpha_2,\ldots,\alpha_K)^T$. Here, $\xi(t)$ is also called as the eigenfunction. Combining (19), (20) and (23), the objective function (22) of FDDA can be simplified as

$$\begin{aligned} \max_{\xi}\ J_f(\xi) &= \frac{\int\int\xi(t)S_b(t,s)\xi(s)dtds}{\int\int\xi(t)S_t(t,s)\xi(s)dtds} \\ &= \frac{\int\int\boldsymbol{\alpha}^T\boldsymbol{B}(t)\boldsymbol{B}(t)^T\boldsymbol{S}_b^c\boldsymbol{B}(s)\boldsymbol{B}(s)^T\boldsymbol{\alpha}dtds}{\int\int\boldsymbol{\alpha}^T\boldsymbol{B}(t)\boldsymbol{B}(t)^T\boldsymbol{S}_t^c\boldsymbol{B}(s)\boldsymbol{B}(s)^T\boldsymbol{\alpha}dtds} \\ &= \frac{\boldsymbol{\alpha}^T\int\boldsymbol{B}(t)\boldsymbol{B}(t)^Tdt\boldsymbol{S}_b^c\int\boldsymbol{B}(s)\boldsymbol{B}(s)^Tds\boldsymbol{\alpha}}{\boldsymbol{\alpha}^T\int\boldsymbol{B}(t)\boldsymbol{B}(t)^Tdt\boldsymbol{S}_t^c\int\boldsymbol{B}(s)\boldsymbol{B}(s)^Tds\boldsymbol{\alpha}} \\ &= \frac{\boldsymbol{\alpha}^T\boldsymbol{\Psi}\boldsymbol{S}_b^c\boldsymbol{\Psi}\boldsymbol{\alpha}}{\boldsymbol{\alpha}^T\boldsymbol{\Psi}\boldsymbol{S}_t^c\boldsymbol{\Psi}\boldsymbol{\alpha}} \end{aligned} \tag{24}$$

where $\boldsymbol{\Psi}$ is the correlation matrix of the B-spline basis system, which is also defined by the inner product in (11). Its expanded form is

$$\begin{aligned} \boldsymbol{\Psi} &= \int\boldsymbol{B}(t)\boldsymbol{B}(t)^Tdt=\int\boldsymbol{B}(s)\boldsymbol{B}(s)^Tds \\ &= \begin{pmatrix} <B_{1,p},B_{1,p}> & \cdots & <B_{1,p},B_{K,p}> \\ \vdots & \ddots & \vdots \\ <B_{K,p},B_{1,p}> & \cdots & <B_{K,p},B_{K,p}> \end{pmatrix}. \end{aligned} \tag{25}$$

Obviously, $\boldsymbol{\Psi}$ is a symmetric matrix. Once the B-spline basis system is selected, $\boldsymbol{\Psi}$ can be uniquely determined. Hence, the objective function of FDDA is transformed into an optimization problem like LDA, which can be easily solved in the same method. The optimal solutions of $\boldsymbol{\alpha}$ are the eigenvectors corresponding to the nonzero eigenvalues of the following eigen-problem:

$$\boldsymbol{\Psi}\boldsymbol{S}_b^c\boldsymbol{\Psi}\boldsymbol{\alpha}=\lambda\boldsymbol{\Psi}\boldsymbol{S}_t^c\boldsymbol{\Psi}\boldsymbol{\alpha}. \tag{26}$$

After solving the optimal solution $\hat{\boldsymbol{\alpha}}_j$, we can obtain the corresponding eigenfunctions $\hat{\xi}_j(t)$

$$\hat{\xi}_j(t)=\hat{\boldsymbol{\alpha}}_j^T\boldsymbol{B}(t)=\boldsymbol{B}(t)^T\hat{\boldsymbol{\alpha}}_j,\quad j=1,2,\ldots,C-1. \tag{27}$$

Under the integral transform of the $C-1$ eigenfunctions, we can get the corresponding functional discriminant features of the spectral curve $x_i(t)$, that is,

$$\begin{aligned} \boldsymbol{f}_i &= \int\hat{\boldsymbol{\xi}}(t)x_i(t)dt=\int\boldsymbol{\Lambda}^T\boldsymbol{B}(t)\boldsymbol{B}(t)^T\hat{\boldsymbol{c}}_idt \\ &= \boldsymbol{\Lambda}^T\boldsymbol{\Psi}\hat{\boldsymbol{c}}_i,\quad i=1,2,\ldots,n \end{aligned} \tag{28}$$

where $\hat{\boldsymbol{\xi}}=(\hat{\xi}_1(t),\ldots,\hat{\xi}_{C-1}(t))^T$ and $\boldsymbol{\Lambda}=(\hat{\boldsymbol{\alpha}}_1,\ldots,\hat{\boldsymbol{\alpha}}_{C-1})$. Thus, by selecting $C-1$ eigenfunctions like LDA, we can extract the functional discriminant features $\boldsymbol{f}_i$ from each spectral curve $x_i(t)$ in the function space. Based on the discriminant features, we can effectively implement the classification tasks of HSIs.

B. Framework of Spectral–Spatial Classification

The spatial information can improve the classification effect. In order to fully show the effectiveness of the proposed method, we design a novel framework of spectral–spatial classification based on the RWFM and FDDA. For maintaining the function characteristic of the spatial information, the spatial information is extracted by simply taking the mean in a predefined neighborhood system in this framework. The general overview of the work flow for the proposed spectral–spatial

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Algorithm 1 Functional Feature Extraction by RWFM and FDDA

Input: $\{x_i\}_{i=1}^N$, $\{B_{k,3}(t)\}_{k=1}^K$.
Output: The functional discriminant features $\{f_i\}_{i=1}^N$.

$x_i(t) = c_i^T B(t) + \varepsilon_i(t)$.
for $\lambda = a : b$ **do**
  for $i = 1 : N$ **do**
    $RWS_e(i,\lambda) = (x_i - Bc_i)^T W_i (x_i - Bc_i) + \lambda c_i^T R c_i$.
  end for
  $SE_\lambda = \sum_{i=1}^N (RWS_e(i,\lambda))$.
end for
$\hat{\lambda} = \arg\min_{\lambda\in[a,b]} SE_\lambda$.
for $i = 1 : N$ **do**
  $\hat{c}_i = (B^T W_i B + \hat{\lambda} R)^{-1} B W_i x_i$.
  $x_i(t) = B(t)^T (B^T W_i B + \hat{\lambda} R)^{-1} B W_i x_i$.
end for
$S_t^c = \sum_{i=1}^n (\hat{c}_i - \mu_c)(\hat{c}_i - \mu_c)^T$,
$S_b^c = \sum_{k=1}^C (\mu_c^k - \mu_c)(\mu_c^k - \mu_c)^T$.
Solve the eigen-problem in the Equation (26) and obtain the eigenfunctions $\{\hat{\xi}_j(t)\}_{j=1}^{C-1}$.
Obtain the functional discriminant features $\{f_i\}_{i=1}^N$ based on the Equation (28).
return $\{f_i\}_{i=1}^N$.

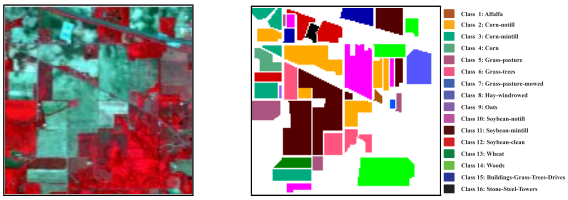


Fig. 1. Flowchart of the proposed spectral–spatial classification method based on the RWFM and FDDA for HSIs.

classification algorithm based on the RWFM and FDDA is shown in Fig. 1. First, the RWFM is used for discovering the function characteristic of the spectral and spatial information (SS) and obtaining their functional curves. Second, FDDA is proposed for extracting the functional discriminant features of the curves. Finally, the fusion features obtained by the concatenation of spectral and spatial features are classified by SVM. The algorithmic procedure of the RWFM and FDDA is given in *Algorithm 1*.

IV. Experimental Results

To demonstrate the effectiveness of the proposed method FDDA + SVM, several vector data processing methods and classical functional data processing methods were used for comparison, such as SS-SVM, FSVM, PCA + SVM, FPCA + SVM [37], LDA + SVM, where SS-SVM denotes a method that directly classifies the fusion of the SS by SVM, and FSVM is one that implements the classification tasks based on functional representations of the SS by SVM. In addition, in order to further demonstrate the advantages of the proposed method, several state-of-the-art methods, i.e., robust joint sparse representation method for handling the band outliers (RJSRB) in [7], PCBSO + M1DI in [19] and PCA + edge preserving features (EPFs) in [22], were selected for further comparison.



(a) (b)

Fig. 2. Indian Pines. (a) False color image. (b) Ground reference map.

A. Experimental Design and Parameter Selection

In these experiments, the proposed method was compared with other methods when optimal parameter setting was used. Unless particularly stated, we randomly selected 10% of the labeled samples for each class as the training set from an HSI, and the remaining 90% as the test set. In the RWFM, the B-spline order p is set to 3. The smoothing parameter λ is tuned in the range $\{10^{-8}, 10^{-7}, \ldots, 1\}$. For the classifier SVM, the penalty term c and the width of the Gaussian kernel σ need to be tuned in the range $\{2^{-10}, 2^{-9}, \ldots, 2^{10}\}$. These optimal parameters were obtained by using a cross-validation technique. In addition, the overall accuracy (OA [%]), average accuracy (AA [%]), and Kappa statistic (κ) in each HSI data set were computed by taking the mean of ten runs. The experiments were implemented using MATLAB R2016a in a PC with Intel(R) Core(TM) i3-4160 CPU and 4G DDR3 RAM.

B. Results on the Indian Pines Data Set

The image gathered by the AVIRIS sensor consists of 145×145 pixels and 200 spectral bands. The ground reference map available is designated into 16 classes ranging from 20 to 2455 in size (see Fig. 2 for details).

To demonstrate the advantage of the proposed model, we compared polylines plotted by spectral vectors with spectral curves after functional data fitting of one sample from each class on the Indian Pines data set in Fig. 3. These results demonstrate that the proposed model can guarantee the smoothness of curves, and obtain a very good fit.

Table I shows the classification results for different methods, including the individual class accuracy, OA, AA, and κ. The proposed method FDDA + SVM achieves the best classification performance in terms of OA, AA, and κ. In addition, FDDA + SVM also achieves the best classification accuracy for most classes. Furthermore, our method

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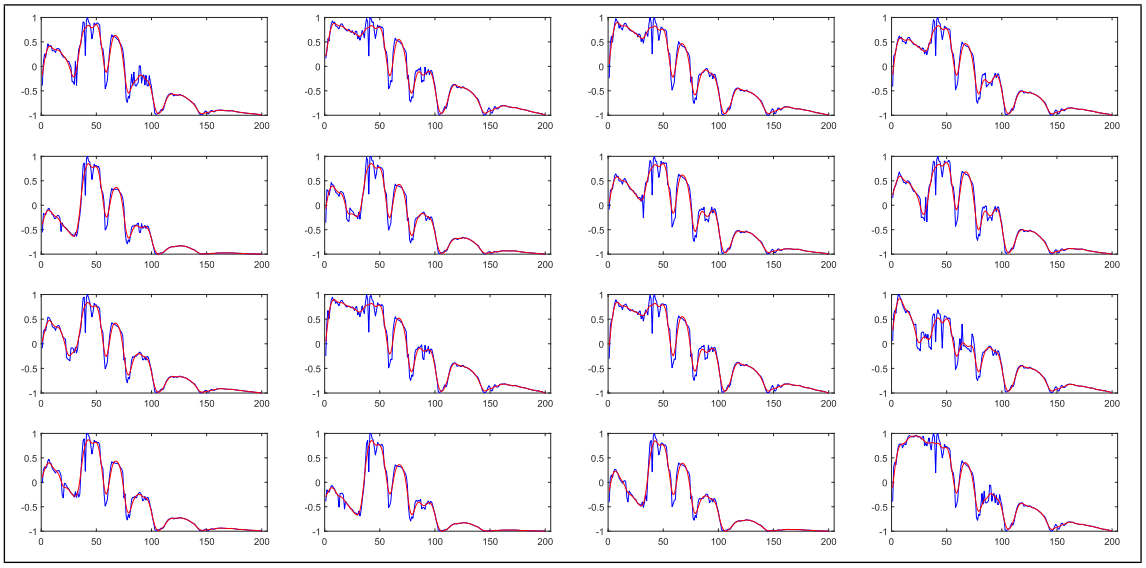


Fig. 3. Polylines plotted by spectral vectors (blue) and spectral curves after functional data fitting (red) of one sample from each class on the Indian Pines data set (16 classes).

TABLE I

CLASSIFICATION RESULTS ON THE INDIAN PINES DATA SET FOR DIFFERENT METHODS WITH TEN LABELED SAMPLES

Class	SS-SVM	FSVM	PCA+SVM	FPCA+SVM	LDA+SVM	FDDA+SVM
1	53.66	56.10	75.61	63.41	**82.93**	75.61
2	81.56	84.90	87.55	92.53	87.78	**96.81**
3	76.44	83.13	85.01	86.08	89.96	**94.51**
4	59.15	79.34	69.48	86.38	85.45	**97.18**
5	92.63	92.40	93.78	94.70	**97.70**	94.47
6	98.78	98.48	99.54	96.50	98.33	**100.00**
7	76.00	80.00	92.00	92.00	**100.00**	**100.00**
8	99.30	99.77	98.37	99.07	**100.00**	**100.00**
9	83.33	88.89	88.89	**100.00**	77.78	83.33
10	79.41	83.87	81.69	82.27	91.19	**95.08**
11	90.22	90.09	90.40	92.94	91.85	**95.20**
12	66.60	69.79	79.36	84.43	86.87	**97.00**
13	97.83	97.83	98.91	98.91	98.91	**100.00**
14	93.59	93.32	93.15	95.61	99.47	**99.82**
15	69.16	72.33	74.93	80.12	84.15	**95.68**
16	77.11	85.54	90.36	**100.00**	97.59	91.57
OA	85.38	87.61	88.66	91.20	92.55	**96.59**
AA	80.92	84.74	87.44	90.31	91.87	**94.77**
κ	0.832	0.858	0.871	0.900	0.915	**0.961**

implements the completely correct classification tasks in four classes. On the other hand, comparing the classical methods (SS-SVM, PCA + SVM, and LDA + SVM) with the functional data classification methods (FSVM, FPCA + SVM, and FDDA + SVM), the corresponding functional data classification methods achieve better classification accuracies. The FSVM outperforms SS-SVM in terms of OA, AA, and κ, which illustrates that the spectral curves can reveal the spectral characteristics of land covers and better reflect their small differences than the spectral vectors. Based on this, we can understand why the FPCA + SVM obtains the better classification accuracies than PCA + SVM, and the FDDA + SVM outperforms LDA + SVM in terms of accuracies. In addition, comparing FSVM, FPCA + SVM with FDDA + SVM, we can find that the functional feature extraction method is effective, and the supervised method outperforms the unsupervised one in this experiment.

Fig. 4 displays the classification maps of the whole scene generated by different methods on the Indian Pines data set. The class boundaries of classification maps for SS-SVM and

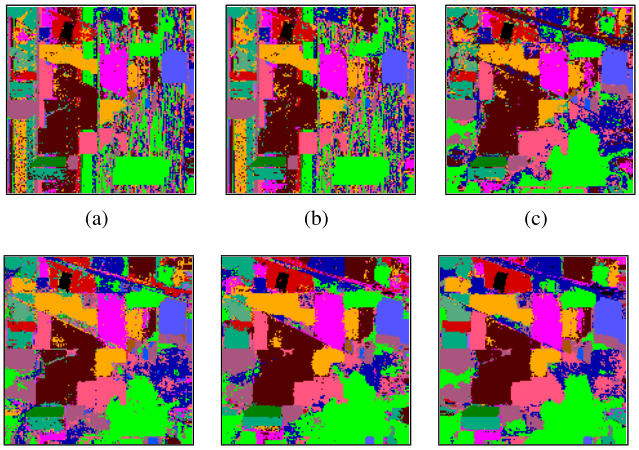
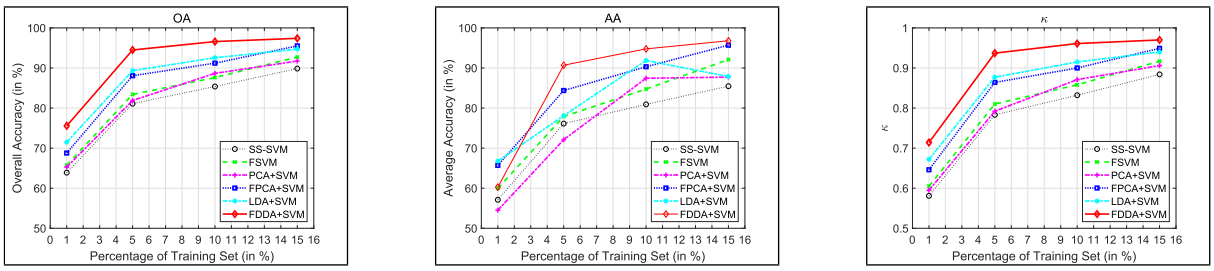


Fig. 4. Classification maps of the whole scene on the Indian pines data set for different methods. (a) SS-SVM. (b) FSVM. (c) PCA + SVM. (d) FPCA + SVM. (e) LDA + SVM. (f) FDDA + SVM.

FSVM were heavily damaged for this data set. By contrast, PCA + SVM and FPCA + SVM achieved a better classification effect than SS-SVM and FSVM in terms of preservation of boundaries. LDA + SVM and FDDA + SVM obtained very good classification maps, which preserved the class boundaries effectively. But compared LDA + SVM with FDDA + SVM, the classification map of FDDA + SVM outperformed the one of LDA + SVM in terms of classification accuracies and preservation of boundaries. Some small classes in the classification map of LDA + SVM were misclassified, which led to the oversmoothing of boundaries. It should be noted that FDDA + SVM kept the boundaries of small classes better.

Table II shows the classification accuracies of RJSRB, PCBSO + M1DI, PCA + EPFs and the proposed method FDDA + SVM on the Indian Pines data set. For PCA in PCBSO + M1DI and PCA + EPFs, we extract features with cumulative eigenvalues of over 99%. However, it should be

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(a) (b) (c)

Fig. 5. Experimental results obtained by different methods when 1%, 5%, 10%, and 15% of samples were selected for each class as the training set of the Indian pines data set. (a) OA. (b) AA. (c) κ.

TABLE II

CLASSIFICATION RESULTS FOR THE STATE-OF-THE-ART METHODS AND THE PROPOSED METHOD ON THE INDIAN PINES DATA SET

Methods	RJSRB	PCBSO+M1DI	PCA+EPFs	FDDA+SVM
OA	96.32	95.52	96.17	**96.59**
AA	86.58	91.85	92.44	**94.77**
κ	0.959	0.949	0.956	**0.961**

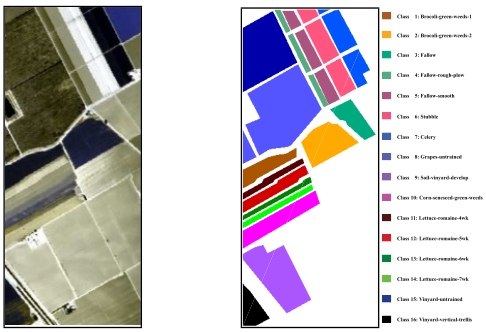
noted that the spatial features in RJSRB, PCBSO + M1DI and PCA + EPFs are extracted by more complex and effective tools than taking the local mean in the proposed method. In particular, in PCA + EPFs, three-layer spatial features are used for classification. Even so, the experimental results given in Table II show that the proposed method outperforms RJSRB, PCBSO + M1DI, and PCA + EPFs in terms of OA, AA, and κ. Thus, the results demonstrate the advantages of the proposed method.

Finally, in order to discuss influence of different training samples to different methods, we randomly selected 1%, 5%, 10%, and 15% of samples for each class in the training set, respectively. The experimental results are displayed in Fig. 5. In the figure, it can be seen as the number of training samples increases, the classification accuracies of almost all methods are improved. The proposed method still achieves the best classification effect in these situations.

C. Results on the Salinas Data Set

This image collected by the 224-band AVIRIS sensor consists of 512×217 pixels. Again, the 20 water absorption bands are discarded. The Salinas ground reference map contains 16 classes (see Fig. 6 for details).

The classification results for the Salinas data set are listed in Table III including the individual class accuracies, OA, AA, and κ. All the methods obtain very good classification accuracies, especially FPCA + SVM, LDA + SVM, and FDDA + SVM, whose OA, AA, and κ are over 99.00%. The proposed method FDDA + SVM performs the best among the methods in terms of accuracies. By contrasting the classical methods (SS-SVM, PCA + SVM, and LDA + SVM) with the functional data classification methods (FSVM, FPCA + SVM, and FDDA + SVM), we can also come to the same conclusions as ones for the Indian Pines hyperspectral data set. In addition, comparing FSVM with PCA + SVM, the spectral



(a) (b)

Fig. 6. Salinas image. (a) False color image. (b) Ground reference map.

TABLE III

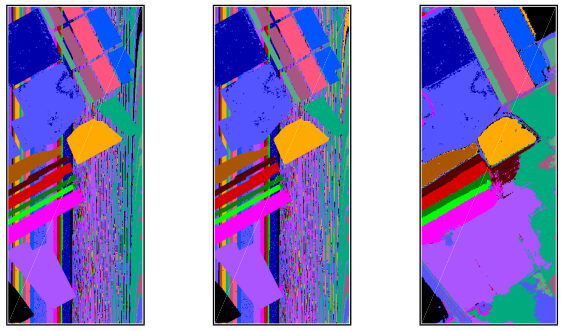
CLASSIFICATION RESULTS ON THE SALINAS DATA SET FOR DIFFERENT METHODS WITH TEN LABELED SAMPLES

Class	SS-SVM	FSVM	PCA+SVM	FPCA+SVM	LDA+SVM	FDDA+SVM
1	99.56	99.94	99.94	99.83	**100.00**	99.94
2	99.67	99.70	99.70	99.67	**100.00**	**100.00**
3	99.78	99.83	**99.89**	99.83	99.89	**100.00**
4	98.88	98.96	99.20	98.80	98.01	**99.52**
5	99.38	99.79	99.75	99.34	**99.83**	99.75
6	99.75	99.78	99.78	**99.94**	99.86	99.89
7	99.75	99.69	99.66	99.53	99.97	**100.00**
8	95.22	96.12	95.68	98.82	98.44	**98.99**
9	99.96	99.96	99.98	99.98	**100.00**	**100.00**
10	98.37	98.24	98.24	99.63	99.93	**99.97**
11	99.90	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**
12	99.94	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**
13	99.39	98.06	98.06	**99.64**	99.15	99.39
14	96.26	98.65	98.65	99.27	98.44	**99.79**
15	92.54	94.10	94.31	97.94	97.13	**98.98**
16	99.51	98.28	98.28	98.89	**100.00**	**100.00**
OA	97.65	98.07	98.01	99.26	99.16	**99.60**
AA	98.62	98.82	98.82	99.44	99.42	**99.76**
κ	0.974	0.979	0.978	0.992	0.991	**0.996**

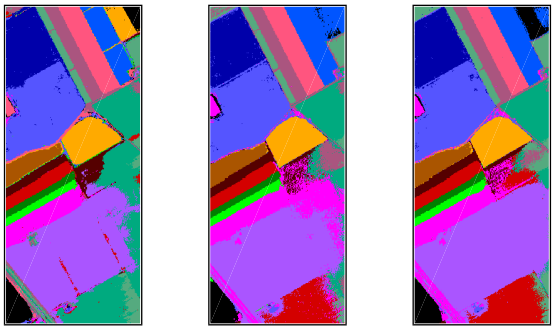
curves can improve the classification in comparison to unsupervised feature extraction PCA for spectral vectors. Comparing FPCA + SVM with LDA + SVM, the unsupervised functional feature extraction FPCA outperforms the supervised feature extraction LDA. This proves the advantage of the function as a tool for processing the HSI in this experiment.

Fig. 7 displays the classification maps of the whole scenes generated by different methods for the Salinas data set. It can be observed from Fig. 6(b) that the spatial distribution of land covers in the Salinas data set is regular, and the spatial resolution of the data is high. Thus, the classification performance is generally better than for the Indian Pines data set. It should be noted that the class boundaries of the classification maps for

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(a) (b) (c)



(d) (e) (f)

Fig. 7. Classification maps of the whole scene on the Salinas data set for different methods. (a) SS-SVM. (b) FSVM. (c) PCA + SVM. (d) FPCA + SVM. (e) LDA + SVM. (f) FDDA + SVM.

TABLE IV

CLASSIFICATION RESULTS ON THE PAVIA UNIVERSITY DATA SET FOR DIFFERENT METHODS WITH TEN LABELED SAMPLES

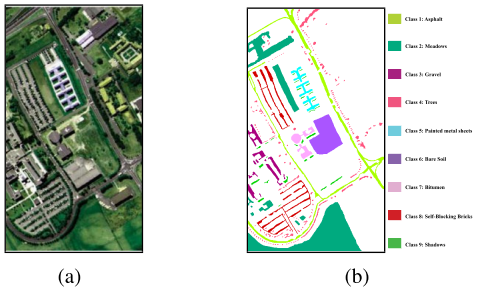
Class	SS-SVM	FSVM	PCA+SVM	FPCA+SVM	LDA+SVM	FDDA+SVM
1	97.92	96.50	98.09	95.79	96.10	**98.29**
2	99.38	99.12	99.52	99.74	99.52	**99.88**
3	57.01	74.43	59.18	90.10	86.66	**98.36**
4	96.12	96.88	96.59	97.46	97.64	**97.82**
5	84.38	87.11	84.63	**99.83**	98.68	99.75
6	93.70	94.41	94.94	98.59	98.34	**99.89**
7	27.82	63.24	27.99	95.91	84.63	**98.16**
8	78.18	87.35	81.07	94.48	90.04	**96.80**
9	7.04	31.81	8.69	94.01	86.27	**96.48**
OA	89.61	92.79	90.28	97.64	96.49	**99.02**
AA	71.28	81.21	72.30	96.21	93.10	**98.38**
κ	0.861	0.904	0.870	0.969	0.953	**0.987**

SS-SVM and FSVM were partially damaged. Visually, other methods achieved better classification maps than SS-SVM and FSVM. In terms of preservations of boundaries, LDA + SVM and FDDA + SVM outperform other methods. Due to the advantages in accuracies, FDDA + SVM can be considered to slightly outperform LDA + SVM.

D. Results on the Pavia University Data Set

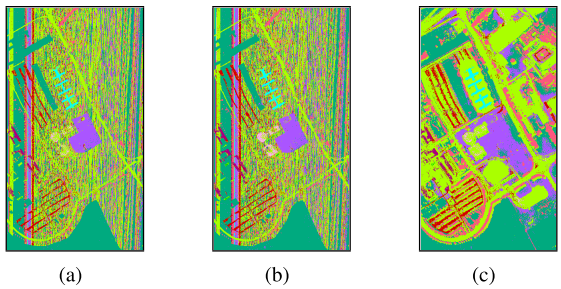
The image acquired by the ROSIS sensor consists of 610 × 340 pixels. A total of 103 spectral bands were used after removing 12 noisy bands. Only nine classes of interest were considered (see Fig. 8 for details).

Table IV shows the classification results for different methods on the Pavia University data set. As can be seen, the proposed method FDDA + SVM performs better than all other compared methods in terms of OA, AA, and κ, respectively, reaching 99.02%, 98.38%, and 0.987. Moreover, the proposed

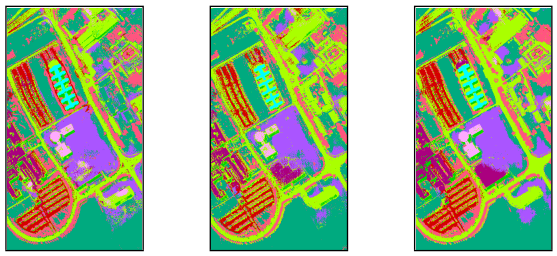


(a) (b)

Fig. 8. Pavia University. (a) False color image. (b) Ground reference map.



(a) (b) (c)



(d) (e) (f)

Fig. 9. Classification maps of the whole scene on the Pavia University data set for different methods. (a) SS-SVM. (b) FSVM. (c) PCA + SVM. (d) FPCA + SVM. (e) LDA + SVM. (f) FDDA + SVM.

method gets the best classification accuracies for almost all classes. This illustrates superiority of the proposed method. Furthermore, FPCA + SVM also obtains excellent classification results, with the OA, AA, and κ, respectively, reaching 97.64%, 96.21%, and 0.969. FPCA + SVM outperforms LDA + SVM, especially in terms of AA. Simultaneously, FSVM is also better in terms of accuracies than PCA + SVM and SS-SVM, giving an improvement of 9.00% on average in terms of AA. This demonstrates that the RWFM and functional feature extraction method are effective and can improve the classification accuracies of the HSI.

Fig. 9 displays the classification maps of the whole scene generated by different methods. Visually, the proposed method FDDA + SVM gives the best classification map. The boundaries of its classification map in Fig. 9(f) are effectively preserved. Although PCA + SVM, FPCA + SVM, and LDA + SVM also obtain very good classification maps in Figs. 9(c) and (e), the boundaries produced by these approaches are somewhat damaged. Worst of all, most class boundaries of the classification maps for SS-SVM and FSVM were heavily damaged. All these maps, visually verify the effectiveness and advantage of the proposed method.

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V. Conclusion

A supervised functional data processing method for HSI classification from a functional point of view has been proposed in this article. The proposed method combines the RWFM with the FDDA method to form a functional data classification framework. First, the spectral vectors are fitted into the spectral curves by the RWFM. The experimental results demonstrate that the spectral curves can improve the classification accuracies, compared with the spectral vectors. Second, the FDDA method is used for extracting the functional discriminant features from the spectral curves. The experimental results show that the FDDA method provides a promising classification performance. In this article, we illustrate the effectiveness and superiority of the proposed method on three real HSI data sets. Moreover, the function as a tool has many advantages for processing HSIs. In addition to handling HSIs, it can also handle ultraspectral images with high spectral resolution. The functional data learning methods have a challenge for processing big remote sensing data. Thus, how to deal with big remote sensing data when using functional data learning methods is one of our future research directions.

Acknowledgment

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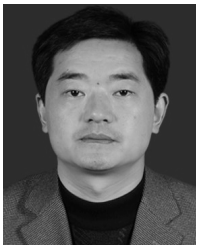
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